

NILE UNIVERSITY RESEARCH INTERNSHIP

TASK 1: CHAOTIC SYSTEMS

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1 Part A: Discrete Chaotic Systems

1.1 Overview

A discrete dynamical system is modeled by applying a fixed map to the state of iteration n to find the state at iteration $n + 1$:

$$x_{n+1} = f(x_n, \text{parameters}) \quad (1)$$

I simulated and analyzed the 1D Logistic Map and the 2D Hénon Map under various control parameters to study stable, periodic, and chaotic behaviors.

1.2 The Logistic Map

The Logistic Map is a 1D nonlinear equation representing population dynamics:

$$x_{n+1} = rx_n(1 - x_n) \quad (2)$$

During the session, we took a plot (which I later learned is called the bifurcation diagram) shown in Figure 1, and I analyzed it. It consists of three parts (single value, periodic, and chaotic), so I will try simulating with one value in each part to compare. When I implemented this diagram myself, it produced the exact same structure as the one from the session, validating my implementation:

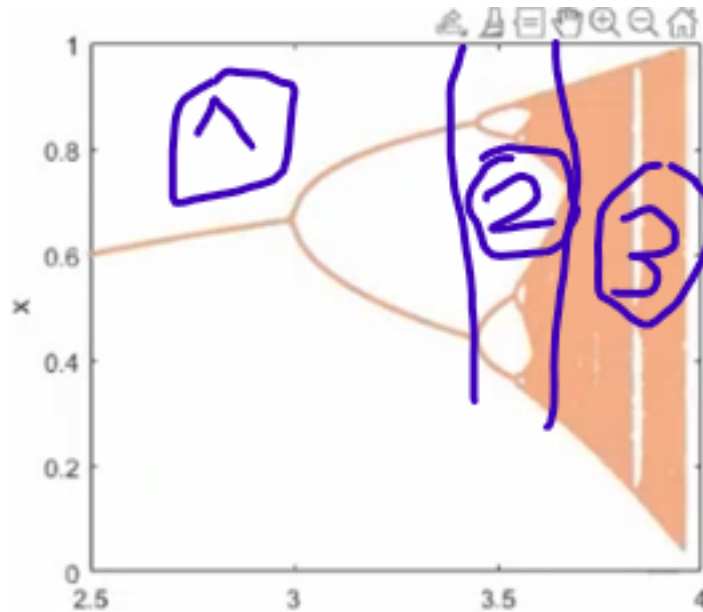


Figure 1: Session bifurcation diagram showing: (1) stable, (2) periodic, and (3) chaotic regions.

What I did:

- Simulated the map for $N = 1000$ iterations ($x_0 = 0.2$), removing the first 500 transient iterations.
- Reconstructed the bifurcation diagram by sweeping the parameter $r \in [2.5, 4.0]$.

Plot Comments & Observations:

- **Stable State ($r = 2.5$):** The system converges to a single fixed point.
- **Periodic Orbit ($r = 3.5$):** The system settles into a periodic oscillation (period-4 cycle).
- **Chaotic State ($r = 3.9$):** The trajectory becomes unpredictable and never repeats.
- **Bifurcation Diagram (Figure 3):** Shows the reconstructed period-doubling cascade and chaotic bands.

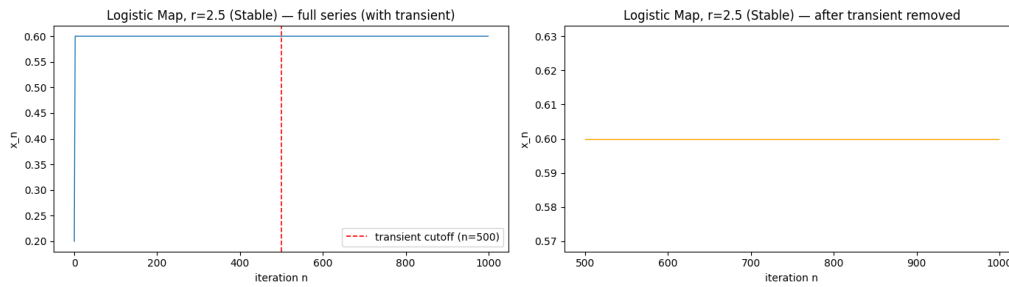
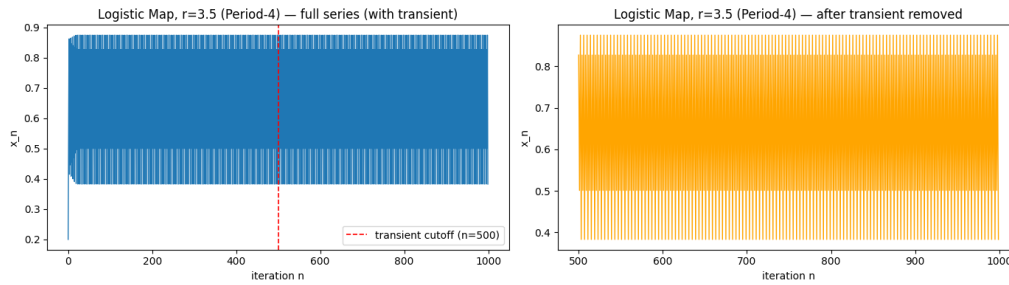
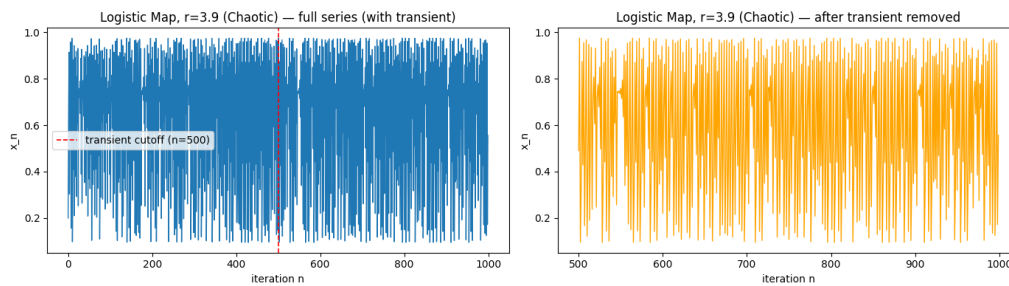
(a) Stable regime ($r = 2.5$).(b) Periodic regime ($r = 3.5$, period-4 cycle).(c) Chaotic regime ($r = 3.9$).

Figure 2: Logistic map time-series simulations before and after discarding transients ($x_0 = 0.2$, $N = 1000$, transient cutoff = 500).

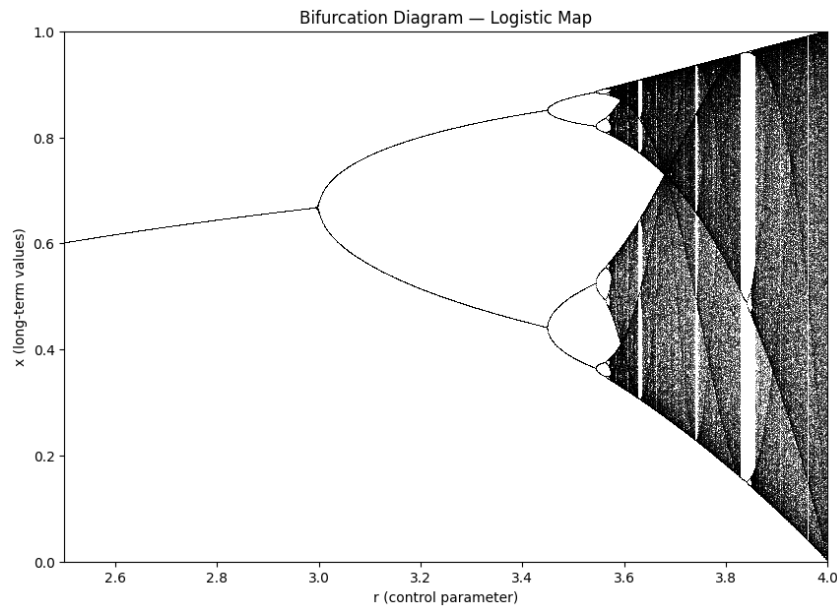


Figure 3: Reconstructed bifurcation diagram of the logistic map.

1.3 The Hénon Map

The Hénon map is a 2D discrete-time system defined as:

$$\begin{aligned} x_{n+1} &= 1 - ax_n^2 + y_n \\ y_{n+1} &= bx_n \end{aligned} \quad (3)$$

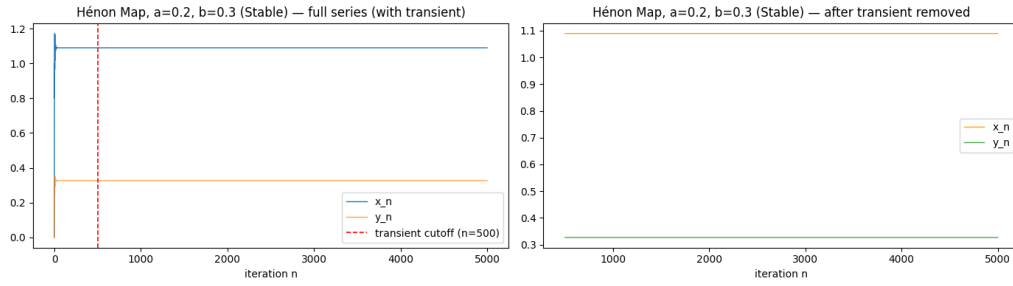
What I did:

- As an extra step (similar to the logistic map analysis), I simulated the map under multiple control parameters to reflect each dynamical state (stable, periodic, and chaotic) with $b = 0.3$ and $x_0 = 0.0, y_0 = 0.0$ over $N = 5000$ iterations.
- Generated the 2D attractor plot by plotting the coordinates (x_n, y_n) after the transient.
- Created the bifurcation diagram by sweeping parameter $a \in [0.0, 1.4]$ while keeping $b = 0.3$.

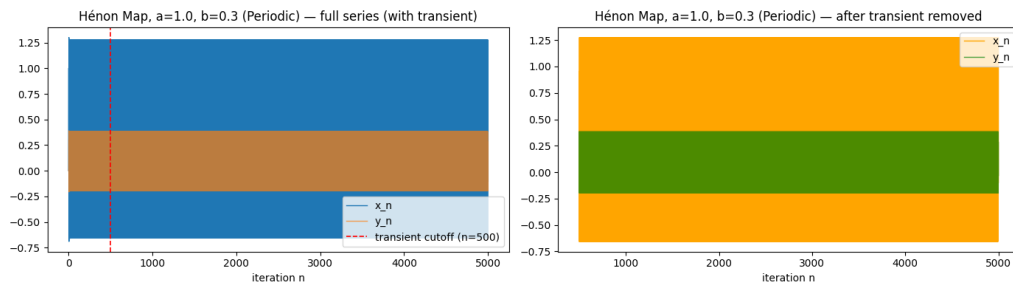
Plot Comments & Observations:

- **Stable State ($a = 0.2$):** The state converges to a single stable coordinate.
- **Periodic Orbit ($a = 1.0$):** The system converges to periodic oscillations.
- **Chaotic State ($a = 1.4$):** The trajectory remains chaotic and does not settle.
- **Attractor Plot (Figure 5):** Since the map is 2D, I plotted x_n vs. its equivalent y_n to construct the strange attractor, showing a folded fractal structure. This plot provides proof that the chaotic behavior is bounded within a specific region and never repeats. This 2D phase-space representation is the main difference from the 1D logistic map.

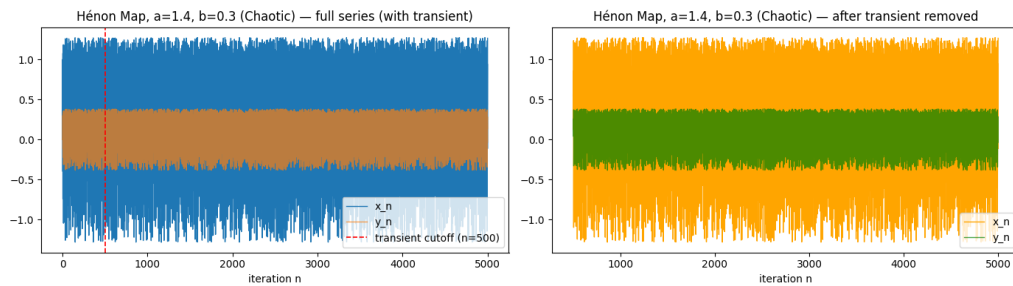
- **Bifurcation Diagram (Figure 6):** Illustrates the period-doubling route to chaos for the 2D map.



(a) Stable regime ($a = 0.2$, $b = 0.3$).



(b) Periodic regime ($a = 1.0$, $b = 0.3$).



(c) Chaotic regime ($a = 1.4$, $b = 0.3$).

Figure 4: Hénon map time-series simulations ($x_0 = 0.0$, $y_0 = 0.0$, $N = 5000$, transient cutoff = 500).

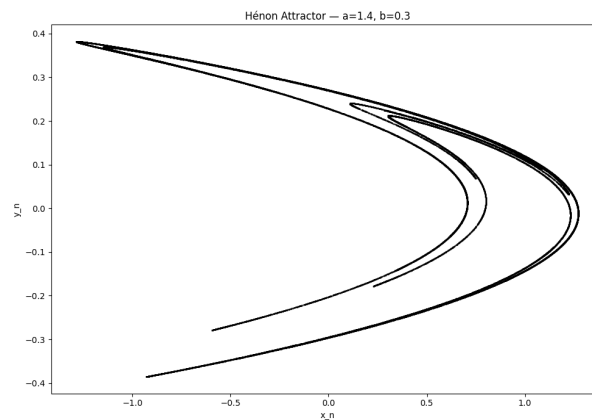


Figure 5: Hénon strange attractor in (x, y) phase space at $a = 1.4$, $b = 0.3$.

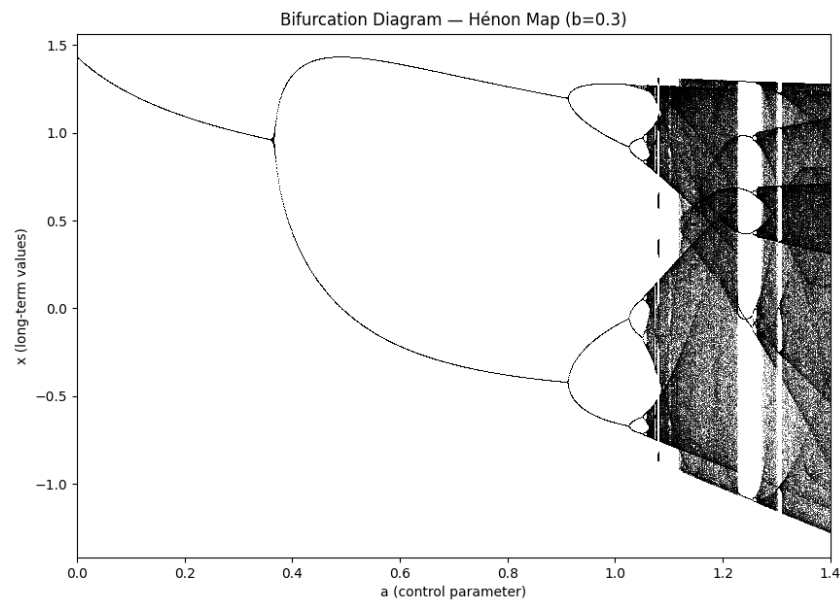


Figure 6: Bifurcation diagram of the Hénon map.

2 Part B: Continuous Chaotic Systems

2.1 Comparison: Chaotic vs. Hyperchaotic Systems

I searched and found that:

- **Chaotic Systems:** Need at least one positive (+), one zero (0), and one negative (−) Lyapunov exponent, which requires a minimum of 3 variables.
- **Hyperchaotic Systems:** Need at least two positive (+), one zero (0), and one negative (−) Lyapunov exponents, which requires a minimum of 4 variables. Hyperchaotic systems are more complex, as shown in the plots.

2.2 Simulation Settings

The simulation parameters and initial conditions (IC) for all continuous systems are consolidated in Table 1 (solved using RK45 via `solve_ivp`):

System	Parameters	Initial Condition (IC)	Time Step (dt)	Length
Lorenz	$\sigma = 10.0, \rho = 28.0, \beta = 8/3$	$[1.0, 1.0, 1.0]$	0.01	50
Rössler	$a = 0.2, b = 0.2, c = 5.7$	$[1.0, 1.0, 0.0]$	0.01	200
Chen	$a = 35.0, b = 3.0, c = 28.0$	$[-10.0, 0.0, 37.0]$	0.001	50
Hyper. Rössler	$a = 0.25, b = 3.0, c = 0.5, d = 0.05$	$[-10.0, -6.0, 0.0, 10.0]$	0.01	500

Table 1: Continuous systems simulation settings.

2.3 The Lorenz System

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z \quad (4)$$

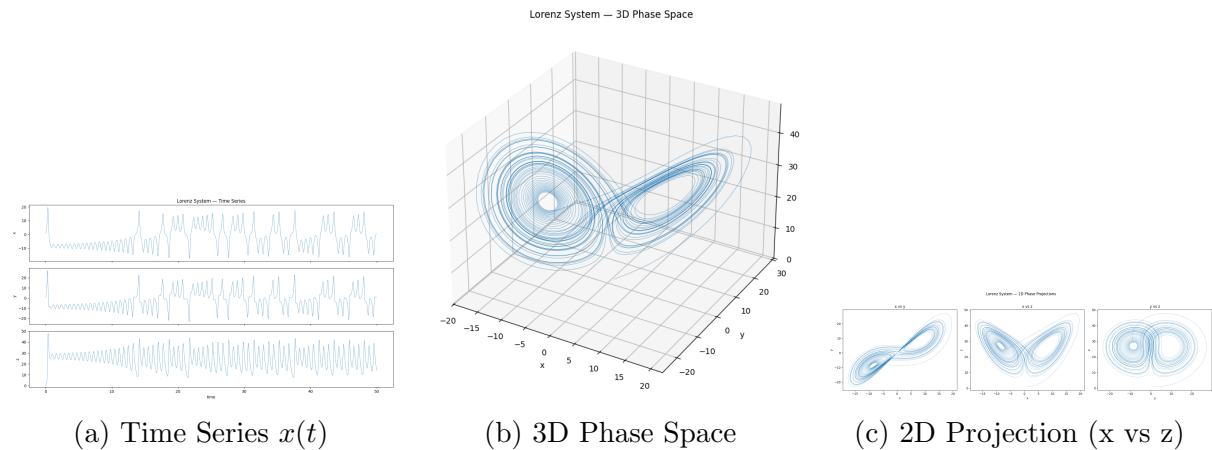


Figure 7: Lorenz system simulation plots.

2.4 The Rössler System

$$\dot{x} = -y - z, \quad \dot{y} = x + ay, \quad \dot{z} = b + z(x - c) \quad (5)$$

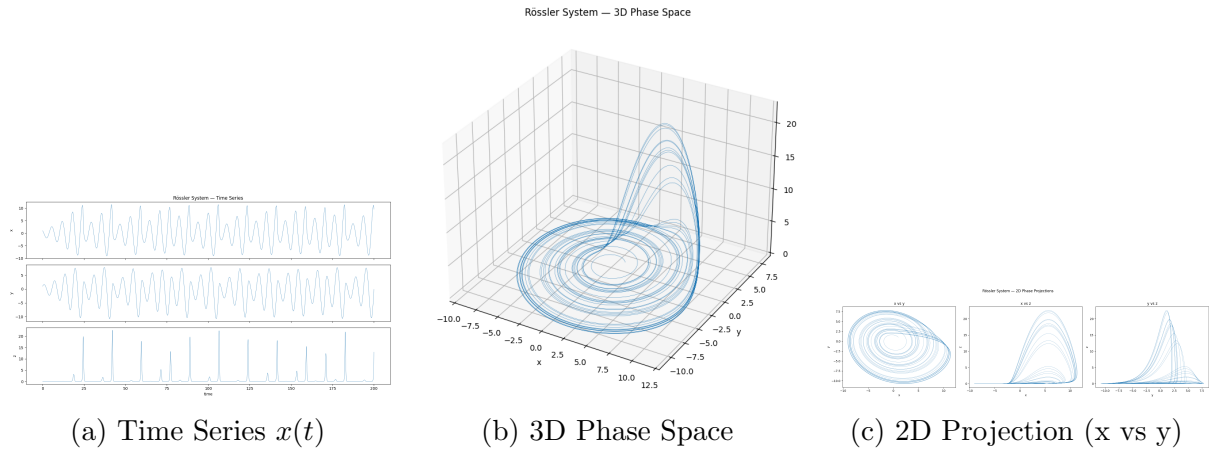


Figure 8: Rössler system simulation plots.

2.5 The Chen System

$$\dot{x} = a(y - x), \quad \dot{y} = (c - a)x - xz + cy, \quad \dot{z} = xy - bz \quad (6)$$

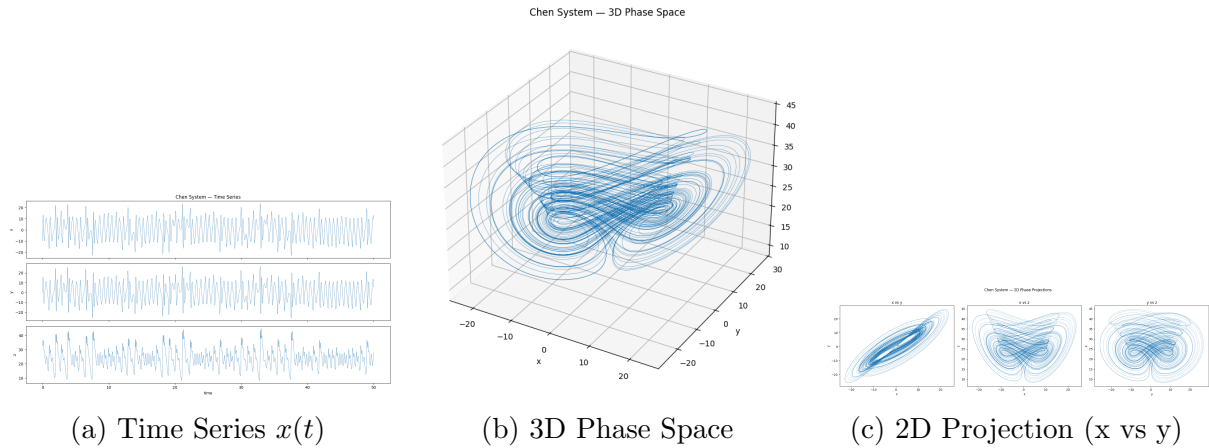


Figure 9: Chen system simulation plots.

Comments & Observations (Lorenz, Rössler, and Chen):

For all three 3D systems (Lorenz, Rössler, and Chen), the plots show irregular oscillations with no repeating period and are bounded, which confirms the characteristics of chaotic systems.

2.6 The Hyperchaotic Rössler System

$$\dot{x} = -y - z, \quad \dot{y} = x + ay + w, \quad \dot{z} = b + xz, \quad \dot{w} = -cz + dw \quad (7)$$

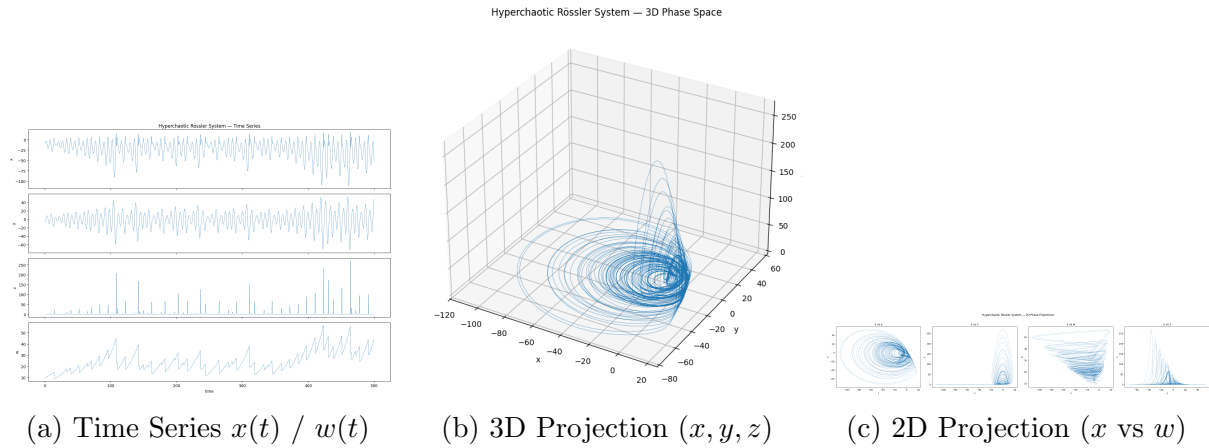


Figure 10: Hyperchaotic Rössler system simulation plots.

Comments & Observations:

The hyperchaotic behavior is more complex, as shown in the plots. The trajectories stretch and separate in multiple directions simultaneously due to the two positive Lyapunov exponents, resulting in a more highly-folded, higher-dimensional state-space structure.

3 Part C: Sensitivity Analysis

3.1 Overview

Chaotic systems are highly sensitive to small errors. Here, I analyze parameter sensitivity and the effect of numerical integration step size (dt) on the chaotic Lorenz system.

3.2 C.1: Parameter Sweep (ρ -sweep)

To study how the system's long-term behavior changes with parameter ρ , I reconstructed a continuous bifurcation diagram (Figure 11). Because the Lorenz system is continuous, I extracted the local maxima of the $z(t)$ trajectory after discarding transients, unlike discrete maps where we take the values directly because they are single values at each iteration.

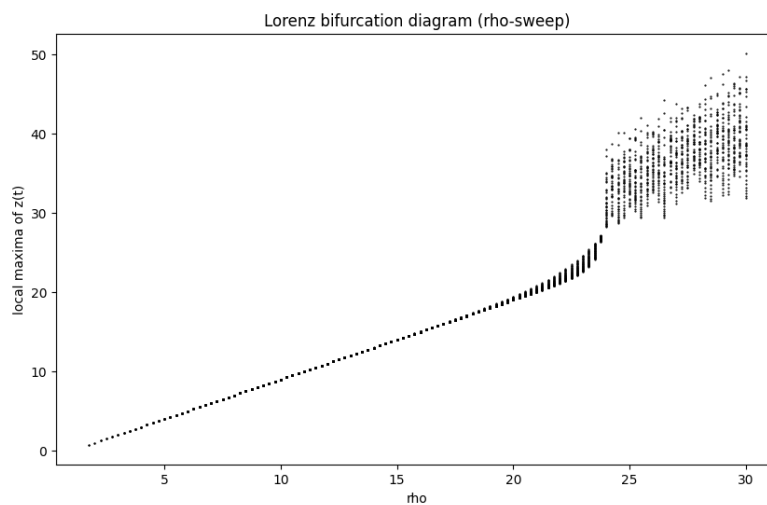


Figure 11: Continuous bifurcation diagram (ρ -sweep) of the Lorenz system.

Comments & Observations:

- **Low ρ ($\rho \leq 15$):** The system is periodic; peaks settle to a single value, producing a single curve.
- **Splitting ($\rho \approx 20 - 24$):** Period-doubling bifurcations occur, causing the curve to split into multiple branches.
- **Onset of Chaos ($\rho \geq 25$):** Trajectories explode into a dense scatter, indicating chaos (consistent with the theoretical threshold of $\rho \approx 24.74$). The empty stripes indicate periodic windows.

3.3 C.2: Step-size Sensitivity

I simulated the Lorenz system using three different step sizes ($dt = 0.05$ (coarse), $dt = 0.01$ (baseline), and $dt = 0.001$ (fine)) to observe how discretization errors accumulate.

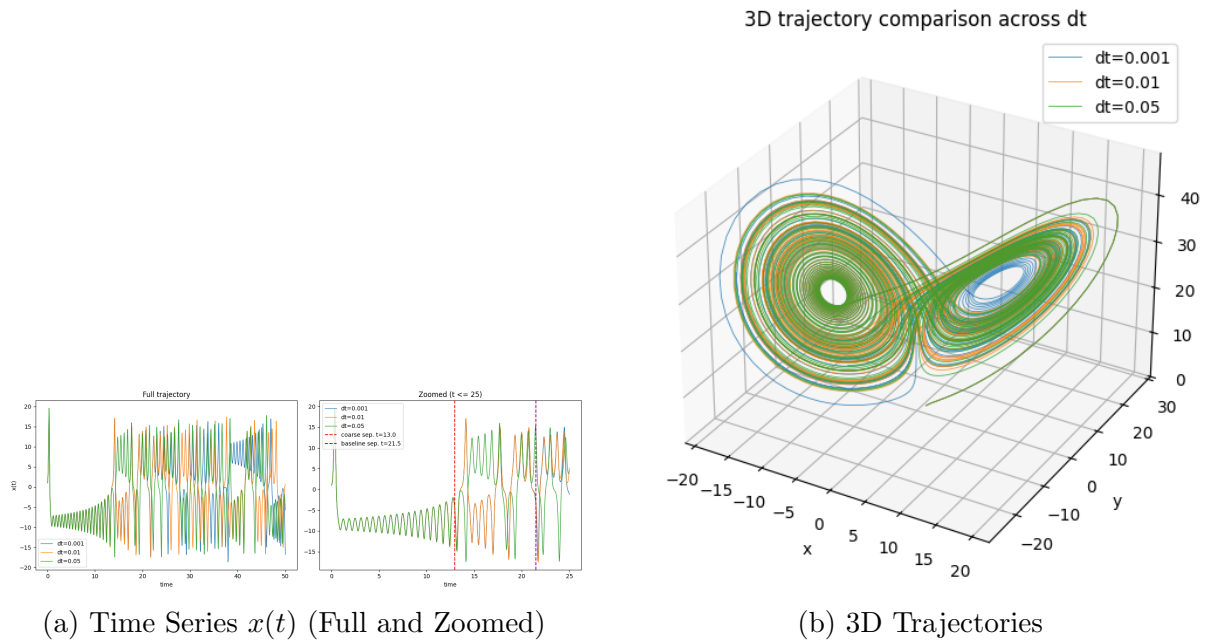


Figure 12: Lorenz trajectory comparisons across step sizes.

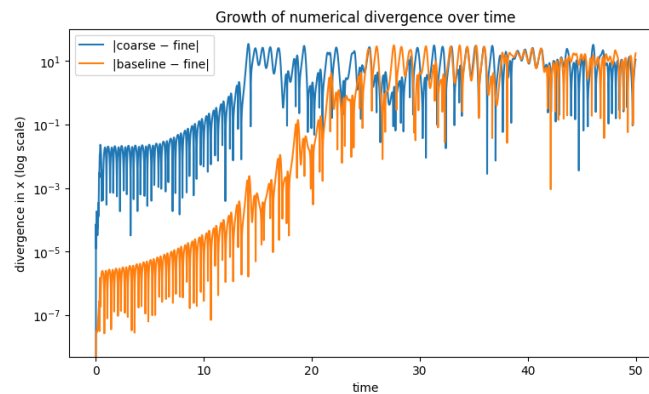


Figure 13: Log-scaled pairwise divergence vs. time.

Comments & Observations:

- Trajectory Divergence (Figure 12a):** Even with identical equations and starting points, the three trajectories separate over time because each dt introduces a different amount of numerical error. The coarse run ($dt = 0.05$) diverges first, at $t \approx 13.0$; the baseline ($dt = 0.01$) follows at $t \approx 21.5$.
- Error Growth (Figure 13):** The difference starts lower in the smaller step size ($dt = 0.01$ vs. fine) compared to the coarser step size ($dt = 0.05$ vs. fine), then over time it explodes and grows roughly exponentially (a straight line on the log scale) until $t \approx 20$, a direct effect of the system's positive Lyapunov exponent. Past $t \approx 25$, the divergence flattens out around 10^1 , since the attractor is bounded and trajectories can't drift apart forever.
- Attractor Robustness (Figure 12b):** Despite following different specific paths, all three trajectories stay on the same butterfly-shaped attractor, showing that the exact path is sensitive to dt , but the overall attractor shape is not.

4 Extra: My Insights from the Session to the Forecasting Problem

Based on what I learned during the session, I connected chaotic theory directly to my time-series forecasting work. The dataset we had in the TCN + LSTM paper was 1D, and through space reconstruction, I realized that we can reconstruct high-dimensional states from just a 1D observation, making it much more rich. I think an LSTM is well-suited for modeling this because of its recurrent memory; by updating its hidden state based on the previous state $h_t = f(h_{t-1}, x_t)$, it naturally acts like a discrete dynamical system that can capture the phase-space transitions. This is an extra reason why the paper of TCN + LSTM achieved such good results. Furthermore, I think we can use the Inception module to model this relation as well, but with multi-scale windows instead of a single fixed window size, allowing the model to adaptively capture different delay scales.